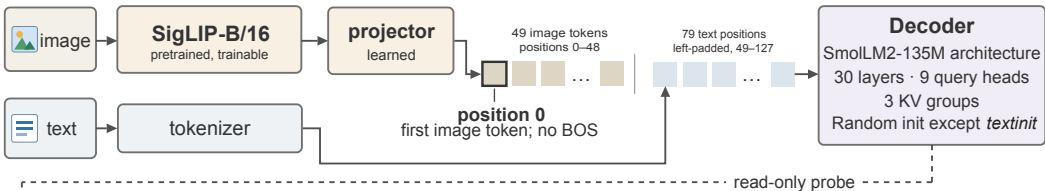


a Model and token sequence



b Measurements

Fixed 32-example probe · every 100 steps

Attention concentration

$$\text{Sink}_l^\varepsilon = \frac{1}{270} \sum_{l,h} \mathbf{1}[a_{l,h} > \varepsilon]$$

a : mean attention to position 0

Value-norm ratio

$$\text{v-ratio} = \frac{1}{30} \sum_{l=1}^{30} \frac{\langle m^{(l)} \rangle_0}{\langle m^{(l)} \rangle_+}$$

m : value norm over 3 KV groups

0: at position 0 · +: pooled over other valid positions

Residual-norm ratio

$$\text{h-ratio} = \frac{1}{30} \sum_{l=1}^{30} \frac{\langle r^{(l)} \rangle_0}{\langle r^{(l)} \rangle_+}$$

r : residual norm at block input

c Training conditions

baseline

Softmax
Random decoder

g1gate

Output-gated softmax
Initial multiplier 0.5

sigmoid

Unnormalized
sigmoid attention

textinit

Softmax
Pretrained decoder

RF

Baseline architecture
Fresh 1B-token run